

Ejer 2 TS1



CASTINEIRA.

Tengo $h(n) = \delta(n) - \delta(n-4)$

$$y(n) = x(n) * h(n)$$

Convolución :

$$y(n) = x(n) * (\delta(n) - \delta(n-4))$$

Distributiva:

$$y(n) = x(n) * \delta(n) - x(n) * \delta(n-4)$$

Como $x(n) * \delta(n) = x(n)$ (Convolution con delta)

$$x(n) * \delta(n-4) = x(n-4) \quad (\text{con delta desplazado})$$

Queda

$$y(n) = x(n) - x(n-4)$$

a) $x(n) = \cos(\omega_0 n T_s)$

$$x(n-4) = \cos(\omega_0 (n-4) T_s) \quad \begin{matrix} \text{ATRASO de 4} \\ \text{muestras} \end{matrix}$$

reemplazo en $y(n) \Rightarrow y(n) = \cos(\omega_0 n T_s) - \cos(\omega_0 (n-4) T_s)$

Expresar como unico coseno

Coseno tipico $\Rightarrow A \cos(\theta + \phi)$

$$\text{Si } \phi = \omega_0 n T_s$$

en el desplazado: $\omega_0 (n-4) T_s = \omega_0 n T_s - 4\omega_0 T_s$

$$y(n) = \cos(\theta) - \cos(\theta - 4\omega_0 T_s)$$

Si uso la identidad trigonométrica:

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$y(n) = -2 \sin\left(\frac{\theta + \theta - 4\omega_0 T_s}{2}\right) \sin\left(\frac{\cancel{\theta} - \cancel{\theta} + 4\omega_0 T_s}{2}\right)$$

$$= -2 \sin\left(\frac{\cancel{\theta} \omega_0 T_s - \cancel{\theta} \omega_0 T_s}{2}\right) \sin(2\omega_0 T_s)$$

$$= -2 \sin(\theta - 2\omega_0 T_s) \sin(\omega_0 T_s)$$

Ahora si $-\sin(\alpha) = \cos\left(\alpha + \frac{\pi}{2}\right)$

$$\omega_0 n T_s - 2 \omega_0 T_s = \alpha$$

$$-\sin(\omega_0 n T_s - 2 \omega_0 T_s) = \cos\left(\omega_0 n T_s - 2 \omega_0 T_s + \frac{\pi}{2}\right)$$

$$y(n) = \overbrace{2 \sin(2 \omega_0 T_s)}^{\text{no dep de } n} \cdot \cos\left(\omega_0 n T_s - 2 \omega_0 T_s + \frac{\pi}{2}\right)$$

Busco " $A \cos(\omega_0 n T_s + \phi)$ " " $\leftarrow$ Δ tempo dep n

$$A = 2 \sin(2 \omega_0 T_s) \quad \text{no dep } n.$$

$$\phi = -2 \omega_0 T_s + \frac{\pi}{2} \quad //$$

$$y(n) = A \cos(\omega_0 n T_s + \phi)$$

$$= \overbrace{2 \sin(2 \omega_0 T_s)}^A \cdot \cos\left(\omega_0 n T_s + \overbrace{\left(\frac{\pi}{2} - 2 \omega_0 T_s\right)}^{\phi}\right)$$

$$(b) \quad x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$x(n-4) = \left(\frac{1}{2}\right)^{n-4} u(n-4)$$

$$y(n) = \left(\frac{1}{2}\right)^n u(n) - \left(\frac{1}{2}\right)^{n-4} u(n-4)$$

$u(n)$ escalon vale 0 $n < 0$ y 1 $n \geq 0$

$u(n)$ escalon 2 vale 0 $n < 4$ y 1 $n \geq 4$

Tenemos 3 zonas

$n < 0$ los 2 escalones = 0 $y(n) = 0$

Para $0 \leq n < 4$

$$y(n) = \left(\frac{1}{2}\right)^n$$

$$n \geq 4 \quad y(n) = \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^{n-4}$$

$\underbrace{\hspace{10em}}_{\text{prop}} = \left(\frac{1}{2}\right)^n \cdot \underbrace{\left(\frac{1}{2}\right)^{-4}}_{16}$

entonces: $n \geq 4$

$$y(n) = \left(\frac{1}{2}\right)^n - 16 \left(\frac{1}{2}\right)^n = -15 \left(\frac{1}{2}\right)^n$$

$$y(n) = \begin{cases} 0 & n < 0 \\ \left(\frac{1}{2}\right)^n & 0 \leq n < 4 \\ -15\left(\frac{1}{2}\right)^n & n \geq 4 \end{cases}$$

© $x(n) = u[n+4] - u[n-2]$

$u(n+1)$ escalon que es 1 $n \geq -1$

$u(n-2)$ escalon que es 1 $n \geq 2$

$x[-1] = 1$
 en $x[0] = 1$
 $x[1] = 1$

} después es 0 porque se cancelan cuando se escribe el z^0 escalon.

Con en $x(n-4)$ se corre todo a $(3, 4, 5)$

$y(-1)$; $y(0)$; $y(1)$ todos 1 porque

$$x(n-4) = 0 \text{ hasta } n=3, 4, 5$$

Como mi $x(n)$ original despues del $n=2$ se hace 0

eso sig $n=3, 4, 5$

$$y(3) = y(4) = y(5) = -1$$

Queda:

$$y(n) = \begin{cases} 1 & -1 \leq n \leq 1 \\ -1 & 3 \leq n \leq 5 \\ 0 & \text{en todo lo demas} \end{cases}$$

Calcular DFT $y(n)$ para $N=8$

$$Y[k] = \sum_{n=0}^7 y(n) e^{-j \frac{2\pi}{8} kn} \quad k=0;1;\dots;7$$

uso el ejercicio c)

$$y(n) = \begin{cases} 1 & -1 \leq n \leq 1 \\ -1 & 3 \leq n \leq 5 \\ 0 & \text{resto} \end{cases}$$

$n = 0, 1, 2, 3, 4, 5, 6, 7$

(Note: In the original image, the values for n=0,1,2,3,4,5,6,7 are written as 1, 1, 0, -1, -1, -1, 0, 0 with red markings above them.)

$$y(n) = [1, 1, 0, -1, -1, -1, 0, 0]$$

$k=0$

$$Y[0] = \sum_{n=0}^7 y(n) \cdot e^{-j \frac{2\pi}{8} (0)n}$$

$$Y[0] = 1 + 1 + 0 - 1 - 1 - 1 + 0 + 0 = -1$$

$k=1$ ya tengo expresion Compleja:

$$Y[1] = \sum_{n=0}^7 y(n) \cdot e^{-j \frac{2\pi}{8} n} = 1 + e^{-j \frac{\pi}{4}} - e^{-j \frac{3\pi}{4}} - e^{-j \pi} - e^{-j \frac{5\pi}{4}}$$

$$e^{-j\pi/4} = \frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2}$$

$$e^{-j\frac{5\pi}{4}} = -\frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2}$$

$$e^{-j\frac{3\pi}{4}} = -\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2}$$

$$e^{-j\pi} = -1$$

$$y[1] = 2 + \frac{3\sqrt{2}}{2} - j \frac{\sqrt{2}}{2}$$